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The Knowledge Production of Ecovillages and the Issues With Contemporary Regenerative Architecture

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Abstract
In January 2024 MOMA NYC featured a large exhibition entitled Emerging Ecologies: Architecture and the Rise of Environmentalism. The exhibition displayed architecture works from the 1970's and the motivation was that their ideas and technologies, while perhaps forgotten, actually held a promise. An important element left out from the exhibition was the work of self-made ecovillages. The aesthetics of self-built ecovillages in earthen and strawbale technologies, have to a large extent been rejected by architectural canons. While architects, engineers and building codes have turned down their applicability due to aesthetic reasons and difficulty in technical assessment, these built experiments still stand soundly and have been carefully maintained by their inhabitants. In the light of the current ecological crises which values and lessons can be learned from +20-year-old ecovillages? This paper examines two cases, a self-built house in an ecovillage and a contemporary regenerative centre designed by architects. The two examples present the difference in ideas and values of what regenerative can mean and do, and to whom.

Keywords**:** *sufficiency; regenerative design; ecological vernacular; ecovillages; architectural history and theory*

1. Introduction

Ecovillages share deep roots with both the Arts and Crafts and the picturesque movements, reflecting a broader historical lineage of resistance to industrialization and a desire to harmonize human habitation with nature. The Arts and Crafts Movement, emerging in the late 19th century as a reaction to the dehumanizing effects of mass production, emphasized craftsmanship, the use of local materials, and a return to traditional building techniques, drawing on the realm of vernacular architecture. These are values closely aligned with the self-built, biogenic architecture seen in many ecovillages. Just as Arts and Crafts advocates like William Morris sought to integrate beauty, utility, and social reform into design, ecovillages often prioritize hand-crafted homes, regenerative materials, and participatory construction processes that empower inhabitants and foster community bonds.

The picturesque movement, with its focus on irregularity, organic forms, and the blending of built environments into natural landscapes, also resonates with ecovillage aesthetics. Rather than adhering to strict architectural formalism,

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ecovillages often adopt asymmetry, curving lines, and the use of raw, unprocessed materials — creating a sense of place that appears "grown" rather than imposed. This connection to the picturesque reflects a shared philosophy: that architecture should work with, rather than against, nature.

While these historical movements and their influential ideas offer a rich and meaningful foundation for ecovillage design, they are often overlooked in contemporary architectural practices focused on regenerative design. While ecovillages constitute a valuable exemplary catalogue of practices and solutions, today, the vision of building in harmony with nature, using sustainable materials, and creating low-tech, energy–efficient homes is just as often associated with a different spatial language that still praises modernist principles of order, cleanliness and straightness. Furthermore, unlike the low-budget work and economic principles that surround ecovillages formation, new regenerative architecture today often requires a high level of investment. The high cost of labour-intensive craftsmanship, the expense of locally sourced materials, and the need for specialized expertise to design and construct buildings that respond to the local climate and environment can create significant financial barriers for architecture. Lastly, an unspoken stigma surrounding the aesthetics of ecovillages within the architectural community often hinders meaningful knowledge exchange between the two spheres.

One of the persons that wrote extensively on regenerative design and founded the Centre for Regenerative Studies in California in 1976 is John Tillman Lyle. He differentiated "development as a matter of economics" from development as the design of a "new ecosystem" were "its structure, function, and spatial distribution of activities determines its effects in terms of both resource use and environmental quality" (Lyle, 1994, pp. 19-25).

Lastly, is important to note that the study draws on Daniel Maudlin's definition of vernacular architecture as "the 'architectural language of the people'" (Maudlin, 2010, p. 13), thus applying a wide understanding of the concept. According to Maudlin, "Vernacular architecture studies can be an inclusive and continuous field of study that can incorporate any number of methodologies and perspectives and which may include such buildings as traditional, regional, anonymous, everyday, modern and contemporary." (Maudlin, 2010, p. 13). Relatedly, ecovillage buildings are examples of 20th century Danish vernacular architecture (Eybye, 2024, p. 500).

2. Objective and methods

This paper examines one house in a Danish ecovillage and one project that was built under regenerative, contemporary ideals. By bringing these two cases together, the objective of this study is to showcase the architectural dichotomies and differences that exist when the 'regenerative' is constructed and owned from different perspectives and by different constituencies.

Methods applied is case study research. Sources comprise literature, site visits, empirical studies, interviews and other relevant material to get critical insights into the aims and visions of the building design. Furthermore, the study draws on theory and history of architecture such as the Arts and Craft Movement, the picturesque, vernacular architecture, regenerative architecture and sustainability.

3. Two regenerative practices in the same landscape – balancing inclusion and elitism

3.1. A house in Hjortshøj

Hjortshøj is a cooperative ecovillage hosting more than 300 people in eight housing groups. It is located 15 km north–east of Aarhus and was founded in 1986, although the first group of inhabitants moved in in 1992.

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The aim of the ecovillage was and still is to build a local community with an organic social cohesion and thus, one which can test for new environmental initiatives and new forms of social collaborations providing economic freedom for many. Most materials were chosen in relation to the climatic function they needed to respond to, including solar orientation, thermal comfort and natural ventilation. The irregular expression of the built works can be understood through the theoretical framework of the Picturesque, which sustained that architecture should feel like nature, much more irregular and diverse than Renaissance geometric articulations (Hussey, 1967/2020). The climatic exchange between inside and outside in Hjortshøj is much more varied and consists of far more 'participants' than in the modernist sealed glass building with mechanical climate controls (Bjerregaard, 2005).

Not only are associative references to the local farmhouse incorporated in relation to the site, but the area is characterized by actual elements from a traditional farming culture: willow grove (for fuel), small artificial mounds for storing vegetables, and roads and paths that are unpaved and contribute to the fostering of biodiversity. By its very nature, the 'self-grown' should not just be seen as a strategy in a figurative sense, as we know it from the architectural theory of the picturesque, but in Hjortshøj it has a consequence that reveals a great variety and wealth of possibilities. The dissemination lack of unifying and holistic wit contained in buildings from the heyday of the picturesque in the 18th and early 19th centuries and in the later Arts & Crafts movement, is an aspect that also characterizes ecovillages like Hjortshøj.

The Arts and Crafts Movement radicalized how to understand material properties. In the work of Gertrude Jekyll for example, materials almost took on an independent 'personality'. (Watkin, 1982). Building materials in general, were not just from the local brickwork production, but made from clay from the building site itself. Within this ideology, buildings had to be built according to local climate and should be freed from symmetrical cannons. Windows, chimneys and eaves would be designed according to the specific needs and location, an extremely situated positioning of the work to be done (Ruskin, 1925).

Through one of the self-built houses in Hjortshøj, this paper discusses the knowledge production that has been gestating in the context of ecovillages. The house was built in 1993, a time when very few architects - and certainly not the big studios - were working with low-tech, ecological technologies. At first glance, there is a similarity in terms of expression and content to the self-build projects of the temporary hippie-camps of 1968 (e.g. Thylejren) and the more established low-rise movement (e.g. Vandkunsten) (Hoff, 1977). But a significant difference is that the ecological traits, understood as anti-industrialism, anti-technical and anti-scientific have been taken a step further. Instead of burning fossil fuels, the focus was on renewable energy, created in the simplest and most low-tech way possible. The choice of building techniques and materials also carefully consider how to build in a way that is as respectful of nature as possible.

The house is located on a windswept hilltop overlooking the Bay of Aarhus. It is oriented towards the south with a one-sided roof slope to the north. A large facade area is then created to the south and a small one to the north. The house is constructed with thick rammed clay walls as load-bearing elements. It is a semi-detached house for two families. The rammed clay walls support the vertical partition and the beams of the roof, whose massive timbers can be seen in the ceilings on the inside, as would have been devised by Arts and Crafts architects.

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Fig. 1 – *Trompe wall* (Eybye, 2024).

The clay walls stand on a ground plinth of conventional design consisting of cast concrete. The clay walls are exposed in the interior, creating a beautiful surface where the small stone particles of the earth sparkle in the slightly rough and matte earth-colored surface. The floors are made of wood. On the outside of the clay walls, there is a thick layer of paper insulation laid in a heavily dimensioned wooden structure (the builders call it 'half–timbering'). The insulation and 'trusses' are clad with wood on the outside.

The timber cladding begins at the first floor, which in picturesque and Arts and Crafts theories is a strategy to achieve a feeling that the house has 'grown out of the ground like a tree', clay and stone on the ground and timber structures 'growing' up high.

At the center of the lower floor on the south façade, an arch evokes the form of traditional Nubian clay architecture, leaving the earthen wall visibly expressed on the exterior.



Fig. 2 – *Hjortshøj street* (Jensen, 2024).

In this way, the clay wall is exposed in the exterior (Fig. 1). The large opening created by the arch is glass-covered, with heavy mullions. In general, the building's window sections have strong wooden mullions, and thus do not pay homage to glass as a material in its own right, with its science- connotations, but instead adhere to John Soanes strategies of 'hiding' the glass. (Giedion, 1997) (Soane 1815), but can also be understood as a reference to the vernacular mullioned window.

The composting toilets and installations for rainwater collection - the water is led to a central tank for the area - are located behind the north facade of the center section. The composting toilets and rainwater form an important nuance between inside and outside and are slightly 'closer' to nature than the living rooms in the clay squares, whose character of being more 'inside' than the middle section is emphasized by the fact that the wood-burning stove and chimneys are located adjacent to them.

It can be said that the house uses the repertoire of the Romantic architectural program in a somewhat random way: the exposed structures in the interior, the self-grown appearance, the variation in building techniques, the division into a more enclosed part around a wood-burning stove and a more exterior part with a terrace, the mud-composting toilets, the rainwater

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installations, etc. The house is really rich and nuanced in the indoor-outdoor relationships with climate as there is no vapor barrier and the clay wall's storage of heat and cold gives it a special dimension in terms of cooperating with the natural climate.

The house has been built as if it has grown from its site - literally through the local clay and figuratively through its weathered expression. It is both foreign to and typical of the place. The house does not live up to the formal modernist rules of minimalist architecture. Neither does not live up to the ideals of either Leon Batttista Alberti or Ludwig Mies van der Rohe. However, it could have been built by one of the English eccentrics who made up the pioneer class of the picturesque.

As a whole, the area is interesting, with its dense, curving village streets (always cobbled, unfortified), its residential greenhouse streets, its barns, sheds and animal husbandry, farming - and not least its string of picturesque follies and chateaus (self-build villas) (Fig. 2). One can come to love this variety in the context of a picturesque landscape with solid roots in Romanticism - a romantic garden. If a Giedionic, main stream modernist architect had planned the design and the connection between the public housing and the individual houses, the transition would not have been so dramatic and the contrasts not so sharp. There would probably have been a more rational pathway and a clearer delineation of which zones could be used for chickens and which for beetroots etc. But this would have moved this housing area away from the picturesque tradition. In that tradition it was exactly a special pleasure for the owner of a romantic park with a picturesque house if the cattle grazed by the Gothic terrace door. On the other hand, an architect who was more architecturally trained in the history and theory of the values of Romanticism could have achieved something better than what the Hjortshøj Cooperative Society looks like today.

If it really had been built by an architect familiar with the architectural theory and form-repertoire of the Romantic picturesque, 'random' trees would long ago have been planted in carefully selected locations, with the planting playing as important a role as the buildings, probably be placed as dramatically, unexpectedly and randomly as everything else.

3.2. A regenerative centre in Hedeskov

The project Hedeskov Centre for Regenerative Practice was finished in 2024. It was designed by architecture practice Djernes & Bell and funded privately and with help from Realdania. The project will be monitored by Aalborg University BUILD AAU. Other collaborators include Local Work Studio, Institut for Agroøkologi AU, SoilValues, Skovskolen KU, Kalø Økologiske Landbrugsskole and Dinesen. The project seeks to demonstrate the importance and values of working with nature in the built environment. Strangely, the project is very close to Hjortshøj and to Friland, another ecovillage with more than 20 years of experience in regenerative practices. Sadly, neither the clients nor the architects approached neither of the two communities. Instead, Hedeskov was a collaborative effort between the architects and their clients—owners of a fertility clinic in Aarhus with a longstanding interest in farming—who, in 1978, purchased 180 hectares of land in Rønde.

The project is the transformation of an old school with an explicit aim of showcasing regenerative architecture. Unlike the aesthetic language found in ecovillages, the well-ordered piece of architecture displays details that were tastefully crafted, earth and wood floors perfectly aligned and well made, natural and locally sourced clay finishes on walls and natural stones. The clients proudly express to have worked with the best Danish craftsmen. Even a British expert on clay finishes was hired to teach other Danish workers particular techniques at the construction site. While the school needed a new roof, many of the original wood and brick components were reused

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Fig. 3 – *Meeting room at the Hedeskov Centre* (Dehlin, 2024).

in various parts of the design. The project consists mostly of meeting rooms (Fig. 3) and some office space as well as a residency apartment with a double bed and a very generous kitchen. There is an adjacent building connected with the Centre that is the daughter's new private house with some exquisitely made earth floors done by Lasse Koefoed Nielsen. The project cost over 10 million Danish kroner (approximately 135,000 euros).

3.3. Comparative analysis

Unlike the house in Hjortshøj, Hedeskov does not experiment with indoor-outdoor possibilities, neither tries to test new ideas for the discipline of architecture. Lacking a clear theoretical framework, the project primarily aims to replicate the functions of a typical bourgeois building—albeit using natural materials. The high cost of its execution shows the big disconnect with market flows and also the privileged position to have the solvency to still afford it. Unlike the low budget economical exemplar found in ecovillages, Hedeskov while projecting itself as forward thinking, it shows the issues that are stake when the regenerative architecture does not dare to be experimental or nurtured by ambitious ideas of worldmaking.

4. Discussion: The ecological class

These recent types of projects in Denmark begin to show the power and importance of what Bruno Latour and Nikolaj Schultz have called 'the ecological class' (Latour and Schultz, 2023). But unlike the sociologists would want to define it, these projects also show that the ecological class is often still very much connected with the definition of social class.

Expert in regenerative design, Chrisna du Plessis explains that during the 1960's and in face with the 1970's energy crises, there was a strong self-sufficient leftist global movement in the West *"*based on locally available, renewable resources and appropriate, small-scale technology as a conscious attempt to limit both consumption and development*."* (du Plessis, 2012, p. 12). She explains that these efforts, however, while appropriate for small-scale rural and peri-urban contexts, were seen as impractical for application at an urban scale and the realities of urban growth. On the other hand, the romantic approach to traditional agrarian practices and the hostility to urban civilization, modernization and industrialization has also been a subject of eco-fascist positions. While regenerative architecture represents an important shift in how we think about building and environmental stewardship, its implementation is hindered by practical, economic, and regulatory challenges. To make it more widespread, there needs to be increased support in the form of subsidies, clear regulations, and the development of affordable technologies. Ensuring that regenerative architecture is accessible to all communities—not just the wealthy—is crucial for achieving both ecological and social equity. Not without regulatory struggles, ecovillages like Friland or Andelssamfundet Hjortshøj among others in Denmark have been practicing forms of regenerative constructions for many years now but these communities' work is often seen as peripheral and outside architectural expectations.

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Nikolaj Schultz claims that "instead of opposing ecology and economy, we will have to face, theorize, and organize these in the same breath" (Schultz, 2020, p. 308). It seems that in order not to oppose the two, architects must think in forms of innovation, care and development that can be implemented at meaningful and impactful scales. A wealthy client can spend all they want in making the most beautiful project, as shown in the Hedeskov example, but what does it do to a society, to an ecology, to the planet? Philosopher Slavoj Zizek suggest that "All these things can — hopefully — be achieved only through strong and obligatory international cooperation, social control and regulation of agriculture and industry, changes in our basic eating habits (less beef), global health care, etc." (Zizek, 2024). He argues that a "representative political democracy alone will not be sufficient for this task" and that "a much stronger executive power capable of enforcing long-term commitments will have to be combined with local self-organizations of people, as well as with a strong international body capable of overriding the will of dissenting nations" (Zizek, 2024).

5. Findings and conclusion

This paper focused on investigating different values and ideas of regenerative architecture. For the purpose of illustration, the paper presented two cases. The Hjortshøj house is grown from is site, permeated with respect of nature, and built on a low budget. Moreover, the house features a number of (architectural) experiments. In contrast, the Hedeskov Centre is an expensive transformation that draws on regenerative principles, yet with a focus on visual delight and with little will to experiment and challenge. Consequently, the cases highlight the dichotomies and differences within regenerative architecture and thus, show the alignment between the ecological class and social class.

Let us hope that some of the future projects carried out by the elitist communities at Hedeskov, as well as in other regenerative centres emerging in the country can begin to address and implement solutions and alternatives to some of these larger mass-involved questions and issues. A suggestion could be to study the self-built + 20-year-old ecovillages, proven to provide financial freedom, liveability and low environmental impact. However, without picking the low hanging fruit of operating more consciously in the architectural strategies of The Picturesque and Arts and Crafts Movement.

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